

Arguments :-

An argument is a process by which a conclusion is formed from a given set of statements called premises or hypothesis.

Valid argument :-

An argument is said to be valid argument iff the conjunction of the premises implies the conclusion. i.e. the argument which yields a conclusion r from the premises $P_1, P_2, P_3, \dots, P_n$ is valid iff the statement

$(P_1 \wedge P_2 \wedge P_3 \wedge \dots \wedge P_n) \rightarrow r$ is a tautology.

e.g.:- Rule of **Detachment** / Modus Ponens

$$\begin{array}{c} P \\ P \rightarrow Q \\ \hline Q \end{array}$$

Suppose that the statement "If it is a shining day then we will go to the park" and "It is a shining day" are the hypothesis are true, then by Modus Ponens, it conclude as "we will go to the park".

Rules of inferenceModus Ponens :-

$$\begin{array}{l}
 P \\
 P \rightarrow q \\
 \hline
 \therefore q
 \end{array}$$

i.e. $[P \wedge (P \rightarrow q)] \rightarrow q$ will be a tautology.

Modus tollens :-

$$\begin{array}{l}
 \neg q \\
 P \rightarrow q \\
 \hline
 \therefore \neg P
 \end{array}$$

i.e. $[\neg q \wedge (P \rightarrow q)] \rightarrow \neg P$ will be a tautology.

Hypothetical syllogism :-

$$\begin{array}{l}
 P \rightarrow q \\
 q \rightarrow r \\
 \hline
 \therefore P \rightarrow r
 \end{array}$$

i.e. $[(P \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (P \rightarrow r)$ will be a tautology.

Disjunctive syllogism :-

$$\begin{array}{l}
 P \vee q \\
 \neg P \\
 \hline
 \therefore q
 \end{array}$$

i.e. $[(P \vee q) \wedge \neg P] \rightarrow q$ will be a tautology. *Spiral*

Addition :-

$$\begin{array}{c} P \\ \hline \therefore P \vee Q \end{array}$$

i.e. $P \rightarrow (P \vee Q)$ is a tautology.Resolution :-

$$\begin{array}{c} P \vee Q \\ \neg P \vee R \\ \hline \therefore Q \vee R \end{array}$$

Simplification :-

$$\begin{array}{c} P \wedge Q \\ \hline \therefore P \end{array}$$

i.e. $(P \wedge Q) \rightarrow P$ will be a tautology.Conjunction :-

$$\begin{array}{c} P \\ Q \\ \hline \therefore P \wedge Q \end{array}$$

i.e. $(P \wedge Q) \rightarrow (P \wedge Q)$ is tautology.

$$\begin{array}{c}
 P \\
 P \rightarrow q \\
 q \rightarrow r \\
 \hline
 r
 \end{array}$$

Date

e.g:- Show that the following argument is valid.

Soln:-

Method-1

$$f = [P \wedge (P \rightarrow q) \wedge (q \rightarrow r)] \rightarrow r$$

Using truth table:

P	q	r	$P \rightarrow q$	$q \rightarrow r$	$P \wedge (P \rightarrow q)$	$P \wedge (P \rightarrow q) \wedge (q \rightarrow r)$	f
T	T	T	T	T	T	T	T
T	T	F	T	F	T	F	T
T	F	T	F	T	F	F	T
T	F	F	F	T	F	F	T
F	T	T	T	T	F	F	T
F	T	F	T	F	F	F	T
F	F	T	T	T	F	F	T
F	F	F	T	T	F	F	T

Since $[P \wedge (P \rightarrow q) \wedge (q \rightarrow r)] \rightarrow r$ is a tautology, the argument is valid.

Method-2

$$[P \wedge (P \rightarrow q) \wedge (q \rightarrow r)] \rightarrow r$$

$$\equiv \neg [P \wedge (P \rightarrow q) \wedge (q \rightarrow r)] \vee r$$

$$\equiv \neg P \vee (P \wedge \neg q) \vee (q \wedge \neg r) \vee r$$

$$\equiv (\neg P \vee P) \wedge (\neg P \vee \neg q) \vee (q \wedge \neg r) \vee r$$

$$\equiv T \wedge (\neg P \vee \neg q) \vee r \vee (q \wedge \neg r)$$

$$\equiv (\neg P \vee \neg q) \vee [(r \vee q) \wedge (r \vee \neg r)]$$

$$\equiv (\neg P \vee \neg q) \vee [(r \vee q) \wedge T]$$

$$\equiv (\neg P \vee \neg q) \vee (r \vee q)$$

$$\equiv [(\neg P \vee \neg q) \vee q] \vee r$$

$$\equiv [\neg P \vee (\neg q \vee q)] \vee r$$

$$\equiv \{\neg P \vee T\} \vee r$$

$$\equiv T$$

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Method - 3

P

$P \rightarrow q$

q

(by modus ponens)

$q \rightarrow r$

r

(by modus ponens)

e.g *

Check the validity of the argument

$P \rightarrow q$

$r \rightarrow \neg q$

$P \rightarrow \neg r$

Solⁿ :

$P \rightarrow q$

$q \rightarrow \neg r$

$P \rightarrow \neg r$

[$\because r \rightarrow \neg q \equiv q \rightarrow \neg r$

contrapositive]

(Law of syllogism).

e.g *

Given the following statements as premises, all referring to an arbitrary meal:

(a) If he takes coffee, he doesn't drink milk.

(b) He eats crackers only if he drink milk.

(c) He does not take soup unless he eats crackers.

(d) At noon today, he had coffee

Whether he took soup at noon today? If so, what is the correct conclusion?

Solⁿ :- Let

P : he takes coffee.

q : he drinks milk.

r : he eats crackers

s : he takes soup.

Then we have,

by condition (a) $P \rightarrow \neg q$

by condition (b) $r \rightarrow q$

by condition (c) $\neg r \rightarrow \neg s$

by condition (d) P

$P \rightarrow \neg q$ a premise

$\neg q \rightarrow \neg r$ (contrapositive of premise (b))

a conclusion $P \rightarrow \neg r$ (by Law of syllogism)

$\neg r \rightarrow \neg s$ a premise

a conclusion $P \rightarrow \neg s$ (by law of syllogism)

P a premise

a conclusion $\neg s$ (by modus ponens)

Hence $\neg s$ is the conclusion, i.e. he did not take soup at noon today.